

Six Birds Foundations V: Endogenous Closure

A Catalog of Structural Laws for Living, Cognitive, and Social Systems

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Abstract

Can a repair be credited to the system that undergoes it, rather than to an observer? This paper states sixteen structural entries, E1–E16, for repairs whose defect, move, budget and audit are records the system carries in its own state. It gives definitions under which such self-repair can be checked, and proves in Lean 4 what can be proved about them; the instances are deliberately minimal. Five entries are laws with a checked instance. A two-state device executes a repair along its own trajectory, and a history record links the reinstatement of its apparatus to that event (E3). A two-level audit tower checks its lower level within a declared record capacity (E2). A clocked device installs a fresh refinement at unboundedly many repair times (E1). A three-state system has an individual containing its own damaged state and repair (E12). Repair transports compose, and the receiver’s obstruction bounds every chain of transported repairs (E13). The general theorems include multiplier conditions that certify a global optimum of a capped allocation, even at a cap (E6), the semilattice of repair joins, a finite argmax for probe acquisition and a budget-reallocation identity. The remaining entries are certificate specifications: a complete certificate yields the stated classification, and supplying it is the open obligation.

Keywords. endogenous closure; emergence calculus; carried record; self-repair; self-maintenance; audit capacity; Karush–Kuhn–Tucker conditions; probe acquisition; certificate specification; Six Birds Theory.

1 Introduction

A cell restores a membrane gradient after a perturbation. A learning agent revises a state abstraction after a failed prediction. A court amends a procedure after a class of disputes has defeated the old one. In each case one may ask a structural question that does not depend on the material: can the repair be credited to the system, or was it supplied from outside? It can be credited to the system if the defect is recorded in the system’s own state, the repair is chosen from the system’s own repertoire and charged to its own budget, the repair step is one of the system’s own transitions, and the result is checked by the system’s own instrument. If any of these records or steps belongs to an experimenter, an engineer or an external administrator, the repair has been supplied.

The earlier papers of this series give exact tools for supplied repair. Foundations IV collects fifty-two structural laws about fixed data: when a readout descends through a quotient, how a failed descent is repaired by refining the source or coarsening the target, when a residual must be priced

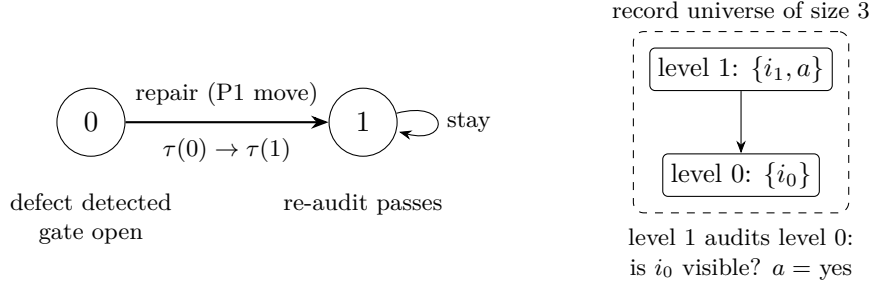


Figure 1. The repair device of Definition 4.1. Left: its trajectory $0, 1, 1, \dots$ takes the lawful repair step $0 \rightarrow 1$ (Theorem 4.4). Right: the two-level audit tower, whose counted records fill a universe of size 3 (Theorem 4.8).

[26]. In all of them the repair is supplied by the theorist: a map fails to descend and the theorist refines its source. This paper asks what the same operations look like when they are coordinates of the system’s own state. The basic move is

$$O \mapsto \text{Carried}_{\mathbb{S}}(O),$$

where a record is *carried* by a system $\mathbb{S}$ if it is a coordinate of the state of $\mathbb{S}$, is updated only by transitions of $\mathbb{S}$, and was produced by a trajectory of $\mathbb{S}$ rather than by a fallback or an external edit (Definition 3.6). Carriedness is a condition on the source of a record. It attributes no purpose to the system.

1.1 A worked example: a repair the system performs

The smallest system that repairs itself has two states (Figure 1). State 0 is damaged and state 1 is repaired; the transition kernel allows a move from either state to 1, and the system’s trajectory is $0, 1, 1, \dots$. The system carries three things in its state. An *instrument* reports a defect exactly when the state is 0, permits a repair of the first structural sort (P1, a lower-operator rewrite) exactly at 0, and accepts a state on re-audit exactly when it is 1. A *ledger* has one budget line. A *repair generator* sends each defect record to a P1 move charged to that line. Every record has the source tag “committed state” and is produced by the system’s own trajectory.

The first step of the trajectory, $0 \rightarrow 1$, is a lawful repair step in the sense of Definition 3.8: the defect is detected, the gate permits the move, the move is charged to the carried ledger and is admissible at 0, the kernel allows $0 \rightarrow 1$, and the re-audit at 1 passes. The system’s history records one challenge episode at time 0 with three entries — the defect, the move and the audit — read at time 1. The record that the system’s apparatus was reinstated at time 0 is linked to that episode, uses its defect and audit, and names $\tau(0)$ and $\tau(1)$ as its source and target (Theorem 4.4). A history with no episode supports no such link.

Give the device two instrument levels. The lower level carries its own instrument record. The upper level checks the lower one — is the lower level’s record among the records its instrument makes visible? — and stores the answer, which is “yes”, as a carried audit record. The records counted at the two levels, $\{i_0\}$ and $\{i_1, a\}$, are disjoint and lie in a record universe of size 3, so the tower’s footprint is at most its capacity, $1 + 2 \leq 3$ (Theorem 4.8).

The example is small on purpose: the defect, move, audit and ledger records of the device each have one value, the tower adds two instrument and two audit values, and formedness and completeness of the inventories are declared. We call such instances *weak* (Section 3.2). What it shows is that the conditions of a self-credited repair — carried defect, generated move, own budget, own transition, own audit, linked history — can be met together by an explicit system and checked.

1.2 How the catalog is sorted

The paper states sixteen entries, E1–E16, over six definitions, D1–D6, and sorts them by what is proved (Table 3). A *law* is a general statement together with an instance in which its hard hypothesis is proved about a concrete system. Five entries qualify: E1 (perpetual self-extension), E2 (bounded reflexivity), E3 (self-maintaining reclosure), E12 (individuation) and E13 (repair transport). A *general theorem* has ordinary hypotheses and nothing open; E6 (priced exposure allocation) and E9 (probe acquisition) are general theorems, and E6 has a numerical instance. The remaining entries are *certificate specifications*: the Lean statement shows that a complete certificate yields a classification, and supplying the certificate for a real system is the open obligation. Two entries, E5 and E8, are best read as definitions with consistency lemmas.

1.3 Main results

Each result below is proved in the text and checked in Lean.

Theorem (E1: perpetual self-extension). *A repair family that recurs beyond every bound, and installs a strict refinement whenever the challenge is new, makes infinitely many strict self-extensions under a challenge that is always new (Proposition 4.16). A clocked device performs a lawful repair at every odd time and installs pairwise distinct refinements that meet both conditions (Theorem 4.17).*

Theorem (E2: audit capacity). *For a declared finite inventory of instrument levels whose counted record sets are pairwise disjoint subsets of a record universe of size $\text{cap}(z)$, the total footprint is at most $\text{cap}(z)$ (Theorem 4.7). The two-level audited tower of the device attains $3 \leq 3$ (Theorem 4.8).*

Theorem (E3: an executed, history-linked reinstatement). *Along the device’s trajectory the step $0 \rightarrow 1$ is a lawful repair, and the reinstatement of its apparatus is linked to a history episode dated at that step whose defect, move and audit entries are carried (Theorem 4.4).*

Theorem (E6: priced allocation). *Given budget and cap multipliers that satisfy stationarity and complementary slackness for a capped linear objective, the allocation is a global optimum, including when it sits at a cap (Theorem 4.11); with weights $(2, 1)$, caps $(1, 5)$ and budget 3 the optimum $(1, 2)$ has a positive cap multiplier (Theorem 4.12). For any supplied multipliers, a selected probe with positive marginal discharge forces a positive price (Theorem 4.9).*

Theorem (E12 and E13: individuals and transport). *Parts that are closed under repair form an intersection-closed family, and integrated individuals are equal or disjoint (Proposition 4.19). In a three-state system the part made of a damaged state and its repair is an integrated individual, while the healthy state alone passes every test except repair closure (Theorem 4.20). Repair transports compose with added payments, and a chain of k transported repairs lowers the receiver’s obstruction by at least k (Theorem 4.23).*

Theorem (E9 and E15: acquisition and reallocation). *Over a finite nonempty list of admissible acquisition candidates that is complete for admissibility, some listed candidate maximizes discharge per cost among all admissible candidates (Theorem 5.1). If an offline phase gates external exchange to zero and the shared budget is exhausted in both phases, the offline internal capacity equals the online internal capacity plus the freed exchange share (Proposition 5.3).*

Theorem (D1 and D2: repair joins and residual traces). *The repair join $q \vee r$ is the greatest lower bound of q and r in the refinement order and is idempotent, commutative and associative up to fibre equivalence (Proposition 3.3). If the adequacy residual satisfies the chain rule under a new probe and the conditional term is positive semidefinite, the residual trace does not increase (Proposition 3.5).*

1.4 Map of results

Table 1 lists the numbered results. Each is proved after its statement, except Proposition 4.5, whose source is given in Table 2.

Table 1. Map of results.

Result	What it establishes
Proposition 3.3	The repair join is, up to equality of fibres, the greatest lower bound in the refinement order, and it is idempotent, commutative and associative.
Proposition 3.5	The conditional surplus is nonnegative exactly when the probe score is at most the quotient score; a positive semidefinite residual has nonnegative trace; under the chain rule with a positive semidefinite subtracted term, adjoining a probe does not increase the residual trace, and a probe $M = BL$ with $K_{DM L} = 0$ leaves it unchanged.
Proposition 4.2	In the repair device, $0 \rightarrow 1$ is a lawful repair step.
Theorem 4.4	In the repair device with its declared trajectory, $0 \rightarrow 1$ is a lawful repair step, the reinstatement record is history-linked with all three entries attached and carried, and the episode is dated at the reinstatement; in a history with no episodes no reinstatement record is history-linked.
Proposition 4.5	A carried self-audit claim whose subject and accepting instrument are the same level, with no bridge to a different level, is not accepted.
Theorem 4.7	For every declared capacity measure the footprints of the declared levels sum to at most the capacity, and a level with the own record of a declared level at a different index is not declared.
Theorem 4.8	The two-level audited tower on the repair device checks its lower level, forms a declared capacity measure with footprint $3 \leq 3$, and keeps the lawful repair step $0 \rightarrow 1$.
Theorem 4.9	For a multiplier witness with positive marginal costs, a selected probe of positive discharge forces a positive price equal to the ratio of every selected probe, unselected probes have no larger ratio, and a slack budget forces price zero.
Proposition 4.10	A multiplier witness at a feasible allocation, for a concave differentiable objective whose partial derivatives are the supplied discharges and with the recorded spend, gives an optimal allocation.
Theorem 4.11	For a capped linear objective on two coordinates, a feasible allocation below its caps with multipliers satisfying the stated conditions is optimal.
Theorem 4.12	With weights $(2, 1)$, caps $(1, 5)$ and budget 3, the allocation $(1, 2)$ is optimal, with one coordinate at its cap; with unit weights it is also optimal.
Proposition 4.16	A repair family with an actual-family forcing certificate, facing a challenge that is new at every time, makes infinitely many strict self-extensions.
Theorem 4.17	The clocked device has a lawful repair step at every odd time and entries beyond every bound; it has an actual-family forcing certificate with a challenge new at every time, so it makes infinitely many strict self-extensions, with distinct refinements at distinct times.
Proposition 4.19	In every individuation setting, intersections of repair-closed parts are repair-closed, and two integrated individuals are equal or disjoint.
Theorem 4.20	On three states, $\{h, d\}$ is an integrated individual containing the damaged state and its repair, while $\{h\}$ is closed under steps and self-maintaining but not a candidate.
Proposition 4.21	Two overlapping maximal boundary regions on three states leave no integrated individual.
Theorem 4.23	Repair transports compose, with payments adding, and along a repair transport a chain of k source repairs lowers the receiver's obstruction by at least k .

Result	What it establishes
Theorem 4.24	An explicit repair transport exchanging two contexts lowers the receiver’s obstruction along a two-step chain, and composing two transports gives one whose payment is the sum.
Proposition 4.25	A paid, role-preserving map that carries every source repair to a receiver repair need not reduce any obstruction.
Theorem 5.1	Over a complete finite nonempty list of admissible acquisitions some listed acquisition maximizes discharge per cost among all admissible ones, also for lists mixing allocation increments and acquisitions, and a policy selecting a listed maximizer dominates every admissible alternative.
Proposition 5.2	Under the chain rule, a probe $M = BL$ with $K_{DM L} = 0$ discharges nothing.
Proposition 5.3	When a shared budget binds and is exhausted in both phases, the offline internal share equals the online internal share plus the external share.
Proposition 5.4	A split pair for the update of a new challenge class through the minimal quotient of a compiled repair blocks descent of its readout.

1.5 What is imported, and what is not claimed

The Karush–Kuhn–Tucker conditions are standard [2]; the chain rule and same-family saturation of the adequacy residual, the split-pair descent criterion, the same-level self-audit obstruction and the source-of-truth tags are taken from earlier papers of the series and are not re-proved here [20, 26, 25]. The paper does not define life, cognition or institutions, and no biological, psychological or social system is shown here to satisfy any law. The instances are small finite systems chosen to make each hard hypothesis provable; the domain readings of Section 7 say which data an application would have to supply, and they are not results.

Roadmap. Section 2 places the paper in the literature, and Section 3 fixes the vocabulary, the six definitions and the catalog. Section 4 states the five laws with their instances, together with E6 and its numerical instance. Section 5 gives the general theorems and Section 6 the certificate specifications and definitions. Section 7 lists inferences the results do not support and the domain readings, and Section 8 records what is formalized. The supplement holds the source concordance, the field lists of every certificate, the full register of non-implications, the finite computations, the synthetic life-facing data, the axiom record and the replay commands.

2 Related work

The Six Birds series. Foundations I defines finite theory packages and the six structural roles P1–P6 [23], and Foundations II supplies admissibility and quotient discipline [24]. Foundations III gives a finite audited interaction calculus: source-of-truth tags, promotion gates, top-down channels, and the theorem that an instrument cannot certify its own soundness at its own level [25]. Foundations IV is the static catalog of fifty-two laws whose split-pair descent criterion and source-refinement repair are the starting point here [26]. Foundations VI returns to an external analyst and studies laws about whole runs [27]; nothing in this paper depends on it. The adequacy residual and its chain rule [20], currencies as constraints and shadow prices [22], hiddenness and exposure [28], the probe-enlargement problem [31], predictive quotients and route transport [29], agents and viability kernels [21], institutions as strict extensions [30] and the reflexive audit loop [32] are used where the laws need them.

Five questions left open in those papers motivate the catalog. Foundations II deferred the case of an instrument whose soundness and completeness are at issue and which is itself part of the system. The reflexive development kept its audit loop on the analyst’s side. The probe-enlargement problem asked which probe to add under which budget. The currency development named attention as a price and the self-regulating accounting cell but did not develop them. The institutions development asked for a carrier that acquires, compiles and reactivates its own repairs. These questions are taken up, in order, by E2; by E1 and E3; by E9; by E6 and E8; and by E4 and E13. None of them is answered in full here.

Biological autonomy. Theories of autopoiesis, closure to efficient causation and organizational closure describe living systems as maintaining the constraints that maintain them [10, 17, 12, 11]. This paper does not identify those theories with its definitions. It takes from them one distinction: the persistence of an object is a different condition from the reinstatement of the apparatus that keeps it viable. E3 states the two as separate certificates. Biological individuality [3, 8], trained immunity [14], allostasis [19] and bioelectric pattern memory [15] appear only as readings that an application would have to support with the data of Section 7.

Regulation, optimization and memory. Ashby’s law of requisite variety and the good regulator theorem are conceptual neighbours of E2 [1, 5]. E6 recovers the Karush–Kuhn–Tucker conditions and the shadow-price reading of a multiplier [2]; rational inattention is a special case when discharge is measured by mutual information [18]. The firm as a boundary chosen to economize on transactions and the Lucas critique are neighbours of E12 and E11 [4, 9]. Reconsolidation after retrieval and offline consolidation are the neuroscience reference points for E14 and E15 [13, 7, 33]. None of these results is re-proved.

Provenance of the results. Table 2 gives the origin of the results that draw on earlier work, with the source and what is added here; the imported tools are listed in Section 1.5.

Table 2. Provenance of the results.

Result	Category	Source and what is new
Proposition 4.5	new; same-level self-soundness theorem imported	The same-level self-audit theorem of Foundations III restricted to a carried level, with the weaker conclusion that the verdict is not <i>accepted</i> [25].
Theorem 4.9, Proposition 4.10, Theorem 4.11	new; KKT conditions and concave sufficiency imported	Theorem 4.9 is the complementary-slackness reading of the Karush–Kuhn–Tucker conditions; the observation that a positive price needs a selected probe of positive discharge, and not only a binding budget, is the one point added here. Proposition 4.10 is their standard sufficiency for concave problems [2, §5.5.3]; it is argued on paper and not formalized. Theorem 4.11 is a separate sufficiency result, proved with cap multipliers, for capped linear objectives, which can fail to be differentiable at a cap, as in Theorem 4.12.
Theorem 5.1, Proposition 5.2	new; saturation specialized	Existence of a maximum over a finite list is elementary; the saturation proposition is the same-family corollary of the adequacy-residual development specialized to the trace, with the chain rule as a hypothesis [20]. The question it answers, which probe to add and at what price, is the probe-enlargement problem [31].

3 Setting

3.1 Packages, quotients and residuals

Definition 3.1 (Theory package). A *theory package* $\mathcal{T} = (Z, f, \Sigma_f, E, \mathcal{A})$ consists of a carrier Z of states; a *lens* f , the map that says what an observer of the package reads off a state; its *definability structure* Σ_f , the family of distinctions expressible through f ; a *completion* E , a map on descriptions that packages a description into a finished object; and an *audit* $\mathcal{A}$, the checks the package applies to its own records. It comes with a transition kernel K on Z whose support $\text{supp } K \subseteq Z \times Z$ lists the allowed one-step moves. The package is *formed* when its completion has a declared fixed point with its records and audit included in the package [23].

The carrier is finite in every application of this paper; in Lean the carrier is an arbitrary type and finiteness is imposed only where a statement uses it. Formedness is a proposition supplied with the package. None of the results below uses the internal structure of f , Σ_f or E .

Quotients and split pairs. A *quotient* is a map $q : X \rightarrow Q$; it records which states an observer can distinguish. Say q *refines* r , written $q \preceq r$, when $q(x) = q(x')$ implies $r(x) = r(x')$. For an update $F : X \rightarrow Y$ and a readout $r : Y \rightarrow R$, the *split-pair obstruction* is

$$\Delta(q, F, r) = \{(x, x') : q(x) = q(x'), r(F(x)) \neq r(F(x'))\}. \quad (1)$$

The readout descends through q exactly when $\Delta(q, F, r) = \emptyset$, and a nonempty obstruction is repaired by refining the source or coarsening the target [26].

Closure deficit and adequacy residual. On a history carrier, let X_t be the history up to time t , let $Y_{t,\tau}^+$ be the declared future observables up to horizon τ , and for a quotient Π of histories let Y_t^Π be the class of X_t under Π . The *closure deficit* $\text{CD}_\tau(\Pi) = I(X_t; Y_{t,\tau}^+ \mid Y_t^\Pi)$ is the information about the future that the history holds beyond what Π keeps [25]. It is used for a current quotient Q and a predictive quotient M refining it. For a covariance operator C , a native probe family L and a dissolving family D , the *adequacy residual* is the Schur complement

$$\Xi_C(D \mid L) = K_{DD} - K_{DL} K_{LL}^\dagger K_{LD}, \quad (2)$$

and adjoining a probe M changes it by the chain rule

$$\Xi_C(D \mid [L; M]) = \Xi_C(D \mid L) - K_{DM|L} K_{MM|L}^\dagger K_{MD|L} \quad (3)$$

[20]. Both identities are imported.

Structural roles. Repairs are typed by the structural roles of Foundations I: P1 lower-operator rewrite, P2 support gating, P3 route reconciliation, P4 staging and refinement, P5 packaging and completion, and P6 audit and accounting. A *repair sort* is one of P1–P5. P6 governs the other moves and is never itself emitted as a repair.

3.2 Laws, general theorems and certificate specifications

Every entry E1–E16 has a general Lean statement and receives one of four statuses.

- A *law* is a general statement together with an instance: a concrete system for which Lean proves the hard hypothesis, so that the conclusion is established for that system. An instance is *weak* when some of its data are declared rather than derived — for example a completeness proposition declared to be true, or a record sort with a single element.
- A *general theorem* has ordinary hypotheses that can be checked system by system, and nothing is left open.
- A *certificate specification*, printed as a Template, is an implication $H \Rightarrow C$ in which H is a named certificate that carries the difficult content, and supplying H for a real system is the *open obligation*. A *conditional classification* is a certificate specification whose conclusion is that exactly one of several statuses holds: given a complete status record, which already asserts that one branch is valid and that every record for the same claim has the same status, the statuses are mutually exclusive. The content is the exclusivity of the branch conditions and the fields each branch requires; the existence of a valid branch is part of the certificate.
- A *definition with consistency lemmas* is an entry whose principal statements follow by unfolding a definition or projecting its fields.

Three words recur in the classifications. A *status record* is a carried record that assigns one status, or *branch*, of a classification to a claim; it is *branch-valid* when the conditions listed for that branch hold for the claim. A residual or defect is *statused* when the ledger carries a record that names it and assigns it a status (unresolved, discounted, charged to a budget line), rather than leaving it unaccounted for.

3.3 The definitional layer

Definition 3.2 (D1: repair join). For quotients $q : X \rightarrow A$ and $r : X \rightarrow B$ on a common carrier, the *repair join* is $(q \vee r)(x) = (q(x), r(x))$. A *source repair* by r replaces q by $q \vee r$.

Proposition 3.3 (repair-join semilattice). For quotients q, r, s, t on X :

1. $q \vee r \preceq q$ and $q \vee r \preceq r$;
2. if $t \preceq q$ and $t \preceq r$, then $t \preceq q \vee r$;
3. $q \vee q$, $r \vee q$ and $q \vee (r \vee s)$ have the same fibres as q , $q \vee r$ and $(q \vee r) \vee s$ respectively;
4. replacing q or r by a quotient with the same fibres does not change the fibres of $q \vee r$.

Thus, up to equality of fibres, $\vee$ is the greatest lower bound in the refinement order, and it is idempotent, commutative and associative.

Proof. All four parts come from the fact that the fibre relation of the join is the conjunction of the two fibre relations. Projection onto the two coordinates gives (1). If t refines q and r , then $t(x) = t(x')$ gives both $q(x) = q(x')$ and $r(x) = r(x')$, hence $(q \vee r)(x) = (q \vee r)(x')$. The fibre relation of $q \vee r$ is the conjunction of the fibre relations of q and r , which gives (3) and (4). $\square$

The join supplies a common refinement. It does not show that a repair occurs, that the system produced it, or that it reduces an obstruction; those are the obligations of E1.

Definition 3.4 (D2: predictive surplus). For a current quotient Q , a predictive baseline M and a horizon τ with $\text{CD}_\tau(M) \leq \text{CD}_\tau(Q)$, the *predictive surplus* is $S_\tau(Q \| M) = \text{CD}_\tau(Q) - \text{CD}_\tau(M)$. In the operator setting the surplus is measured by the trace $\text{tr } \Xi_C(D | L)$.

Proposition 3.5 (conditional surplus inequalities). 1. For rational scores d_Q, d_M , the difference $d_Q - d_M$ is nonnegative if and only if $d_M \leq d_Q$; it is zero when $d_M = d_Q$, and equals d_Q when $d_M = 0$.

2. If $\Xi_C(D \mid L)$ is positive semidefinite, its trace is nonnegative.

3. Suppose the residual Ξ' after adjoining M satisfies the chain rule (3) and the subtracted term is positive semidefinite. Then $\text{tr } \Xi' \leq \text{tr } \Xi_C(D \mid L)$.

4. If moreover $M = BL$ for some matrix B and $K_{DM|L} = 0$, then $\text{tr } \Xi' = \text{tr } \Xi_C(D \mid L)$.

Proof. (1) is arithmetic. (2) and (3) use that the trace is nonnegative and monotone on the positive semidefinite cone; in (3) the chain rule writes Ξ' as $\Xi_C(D \mid L)$ minus a positive semidefinite term. In (4) the subtracted term vanishes because its left factor is zero. $\square$

The proposition is conditional: the score comparison, the chain-rule identity and positive semidefiniteness are hypotheses. Refining a quotient does not by itself make the surplus nonnegative, and scores taken at different horizons or against different future families are not compared.

The central notion of the paper is a record that the system itself holds. Informally, a record is the system's own when it is a coordinate of the system's state, reached along a legitimate trajectory of the system's own kernel within scope, tagged as coming from committed state or from audited records rather than from an outside source, and flagged as generated by the system within scope.

Definition 3.6 (D3: carried record). Let $\mathcal{T}$ have carrier Z and kernel support $\text{supp } K$. A record value ρ with source tag σ , generation flag g and scope flag i is *carried* if

1. $\rho = \rho_{\text{of}}(z)$ for a declared coordinate map $\rho_{\text{of}} : Z \rightarrow R$ and some state z of a declared trajectory τ ;
2. τ starts at a legitimate start and every step up to z is in scope and lies in $\text{supp } K$; and
3. σ is *committed state* or *audited cell records*, and $g = i = \text{true}$.

A *carried instrument* is an instrument whose visibility, threshold and check-rule records are all carried and whose record list is declared complete.

A value from a fallback, a simulation trace, an ablation record or an independent witness is thus not the system's own record, whatever flags it carries (Lemma S3.1).

Definition 3.7 (D4: E-system and lawful repair step). An *E-system* $\mathbb{S} = (\mathcal{T}, I_S, \Lambda_S, \mathfrak{R}_S)$ consists of a formed package $\mathcal{T}$; a carried instrument I_S with predicates *detects*, *gate allows* and *re-audits*; a carried ledger Λ_S , a list of entries declared complete, each carried; a repair generator $\mathfrak{R}_S$ sending each defect record to a repair move of some sort in P1–P5 with a carried move record and a budget line; and an admissibility predicate for moves.

Definition 3.8 (lawful repair step). For states z, z' , a defect record d and an audit record a , a *lawful repair step* $z \rightarrow z'$ requires: d is carried; I_S detects d at z ; the gate allows $\mathfrak{R}_S(d)$ at z ; the move is admissible at z ; its budget line is an entry of Λ_S ; $(z, z') \in \text{supp } K$; I_S re-audits z' with record a ; and a is carried.

Each of detection, gate permission, kernel transition and re-audit is necessary (Lemma S3.2).

The step is a lawful possible transition with linked records. It does not show that the transition takes place along a trajectory, nor recurrence, reduction of an obstruction, feasibility of a family of repairs within the budget, or maintenance over time.

Definition 3.9 (D5: closed-loop scope). A *challenge history* lists challenge episodes and repair-typed audit entries (defects, moves, audits), each entry associated with an episode, together with a proposition declaring the history complete. It is *closed-loop* for $\mathbb{S}$ when it is declared complete and every repair-typed entry in it is carried.

A single listed entry with source tag fallback, or a listed entry with a false generation or scope flag, defeats closed-loop scope (Lemma S3.3). A history with no repair entries is closed-loop. This is intended: closed-loop scope rules out external repair; it does not assert that any repair took place.

Definition 3.10 (D6: probe economy). A *probe economy* has a finite catalog of probes, a carried active family L with support and weights, an exposure budget carried in Λ_S , and three moves: *allocation*, which reweights the current support; *acquisition* of a catalog probe M outside the support; and *retirement* of an active probe. An acquisition is *strict* when M is not same-family saturated with respect to L ; in the operator setting, $M = BL$ with $K_{DM|L} = 0$ witnesses saturation.

A lawful allocation keeps the support, and a lawful acquisition is strict (Lemma S3.4).

3.4 The catalog

Table 3 lists the sixteen entries with their status. The third column says what Lean proves. The fourth names the instance of a law, or the certificate that is the open obligation of a template. The field lists of all certificates are in Supplement Section S5.

Table 3. The catalog.

Entry	Status	Proved in Lean	Instance or open obligation
E1 internalization: a discharged recurring challenge has a carried repair source; perpetual self-extension	law, weak instance	forced self-extension from recurrence and installation; clocked device; five statuses exclusive	Proposition 4.16 and Theorem 4.17; internalization status remains a certificate (Template S4.1)
E2 bounded reflexivity: audit towers are stratified and capacity-bounded	law, weak instance	capacity bound over a declared inventory; two-level audited tower; priority status partition	Theorems 4.7 and 4.8; forced stratification remains a certificate
E3 self-maintaining reclosure: object closure and apparatus reinstatement are separate	law, weak instance	executed repair with history-linked reinstatement	Theorem 4.4; maintained-closure status remains a certificate (Template 6.1)
E4 repair compilation: repeated repair enters lower support	certificate specification	compiled status from a full compilation case; brittleness from a split pair	compilation case (Template S4.2)
E5 reclosure collapse: no admissible carried rescue restores viability	definition	falsifier refutes collapse; seven statuses exclusive	complete rescue inventory (Definition S4.3)
E6 priced exposure allocation (attention)	general theorem	multiplier algebra; sufficiency for capped linear objectives; optimum at a cap	numerical instance Theorem 4.12; an instance on a carried probe economy is open
E7 alarm: a blind spot that threatens viability is brought into current attention, declared an open boundary, or discounted	conditional classification	the three dispositions exclusive	a unique total disposition package for the specified blind spot and horizon (Template S4.5)

Entry	Status	Proved in Lean	Instance or open obligation
E8 control price: carried control signals contain shadow-price components	definition	component and summary lawfulness unfold definitions; seven statuses exclusive	component-lawful case (Definition S4.6)
E9 strict probe acquisition (curiosity)	general theorem	finite argmax; operator saturation gives zero discharge	— (Theorem 5.1)
E10 cognitive demarcation, with intention and goal	conditional classification	statuses exclusive for cognition, intention and goal	complete status records (Template S4.8)
E11 institutional rewrite: a certified rewrite of what a lower layer can do	conditional classification	statuses exclusive; a change of probabilities alone, or a label whose effect fades, is not a rewrite	rewrite comparison (Template S4.9)
E12 individuation: maximal jointly closed subcarrier with a sufficient boundary	law, weak instance	repair-closed parts closed under intersection; individual containing its repair; overlapping maxima	Theorem 4.20 and Proposition 4.21; budget and record closure remain a certificate (Template S4.10)
E13 repair transport: communication, teaching, coercion null, symbolic repair	law (paid transport), weak instance	composition; chain bound; transport across contexts	Theorems 4.23 and 4.24; teaching, coercion and symbolic repair remain certificates (Template 6.2)
E14 reconsolidation: a conflict exposed by recall is repaired, coarsened or recorded as unresolved	conditional classification	lawful trichotomy and nine statuses exclusive	complete outcome inventory (Template S4.12)
E15 offline reclosure: debt discharge alternates in the deficit regime	conditional classification	eight statuses exclusive; reallocation identity	complete status record (Template S4.13); identity in Proposition 5.3
E16 adaptability: routes that agree now but differ in future repair capacity	conditional classification	eight statuses exclusive; agreement now and unequal later repair probabilities read from the record	complete status record (Template S4.14)

4 The laws and their instances

This section states the five entries that are laws and proves their instances, together with the general theorems of E6 and a numerical instance of them. Section 4.1 defines the repair device. Section 4.2 treats E3 and Section 4.3 E2, both on the device; Section 4.4 treats E6, Section 4.5 E1, Section 4.6 E12 and Section 4.7 E13.

4.1 The repair device

Definition 4.1 (repair device). The *repair device* is the E-system with carrier $Z = \{0, 1\}$, kernel support $\text{supp } K = \{(z, 1) : z \in Z\}$ and declared trajectory $\tau(0) = 0$, $\tau(n) = 1$ for $n \geq 1$. Defect, move, audit and ledger records each have a single value, and each is read off the state by a constant coordinate map with source tag *committed state* and both flags true. The instrument detects a defect at z exactly when $z = 0$, allows a repair exactly when $z = 0$, and re-audits z' exactly when $z' = 1$. The ledger has one line. The generator sends the defect to a move of sort P1 charged to that line, and a move is admissible at z exactly when $z = 0$. The package is declared formed, and the instrument and ledger inventories are declared complete.

In Lean the states 0 and 1 are the Booleans `false` and `true`.

Proposition 4.2 (a lawful repair step). *In the repair device, $0 \rightarrow 1$ is a lawful repair step (Definition 3.8).*

Proof. The defect and audit records are carried: each is the value of a declared constant coordinate at a state of the declared trajectory, which starts legitimately and whose steps lie in $\text{supp } K$, and each has an allowed source tag with both flags true. At $z = 0$ the instrument detects the defect, the gate allows the generated P1 move, and the move is admissible; its budget line is the single ledger entry; $(0, 1) \in \text{supp } K$; and the re-audit at 1 passes. $\square$

The declared formedness and inventory completeness are propositions set to true, and the record sorts have one element each; this is why the instances on the device are graded weak.

4.2 E3: self-maintaining reclosure

E3 separates two conditions on a running package $\mathcal{T}_t$ with apparatus App_t :

$$\underbrace{E_t^2 \simeq E_t}_{\text{object closure}} \quad \text{and} \quad \underbrace{\text{App}_{t+1} = \mathbf{m}(z_t) \text{ with a carried, history-linked reinstatement}}_{\text{apparatus reinstatement}}.$$

Object closure says that the formed object is a fixed point of its completion. Reinstatement concerns the *apparatus* — the gates, thresholds, boundary and audit data that keep the object viable. A *maintenance operator* $\mathbf{m}$ maps a state to an apparatus. A *reinstatement record* names a time t , a source and a target state, the operator, the apparatus records before and after, the output record and a ledger entry. The *reinstatement condition* requires that App_t , App_{t+1} and the operator be carried records, and that the reinstatement record be carried and consistent: its operator, apparatus and output fields match, $\text{App}_{t+1} = \mathbf{m}(\text{source})$, its ledger entry lies in Λ_S , and $(\text{source}, \text{target}) \in \text{supp } K$.

Definition 4.3 (history-linked reinstatement). A reinstatement record ρ at time t is *history-linked* in a challenge history H if it satisfies the reinstatement condition and there are an episode e of H , a defect record d , an audit record a and one entry class such that

1. e is dated t and its obstruction witness is d ;
2. the defect entry d , the move entry $\mathfrak{R}_S(d)$ and the audit entry a all belong to H , are all attached to e , and are all carried; and
3. $\text{source}(\rho) \rightarrow \text{target}(\rho)$ is a lawful repair step with defect d and audit a .

Theorem 4.4 (E3: an executed, history-linked reinstatement). *Give the repair device the declared trajectory $\tau(0) = 0$ and $\tau(n) = 1$ for $n \geq 1$. Let H be the history with one episode at time 0, whose obstruction witness is the defect, and three entries — the defect, the move and the audit — attached to it, each read at time 1 with source tag committed state. Let $\mathbf{m}$ return the apparatus of time 1, and let ρ be the reinstatement record at time 0 from state 0 to state 1. Then:*

1. the step $\tau(0) = 0 \rightarrow \tau(1) = 1$ of the trajectory lies in $\text{supp } K$ and is a lawful repair step;
2. ρ is history-linked in H , with all three entries present, attached to the episode and carried;
3. the episode is dated at the time of ρ , and the source and target of ρ are $\tau(0)$ and $\tau(1)$.

Conversely, in a history with no episodes no reinstatement record is history-linked.

Proof. Each part is checked against the definitions: the repair step as in Proposition 4.2, the reinstatement condition field by field, and the dates directly. (1) $\tau(0) = 0$, $\tau(1) = 1$ and $(0, 1) \in \text{supp } K$; the repair step is checked as in Proposition 4.2, with the records carried at time 1 of the trajectory, which starts legitimately and whose only earlier step is $(0, 1)$. (2) The reinstatement condition is checked field by field from the definitions of the apparatus, the operator and the record; the episode and the three entries are listed in H and attached to the episode, and each entry is carried at time 1. (3) The episode time, the time of ρ and the entry time minus one are all 0. The converse holds because condition (1) of Definition 4.3 requires an episode of H . $\square$

This is the instance that makes E3 a law: the repair is a step of the device's own trajectory, and the reinstatement record is linked to the episode of that step, which an empty history cannot supply. The full classification of maintained closure is a certificate specification (Template 6.1).

4.3 E2: bounded reflexivity

E2 limits what a tower of carried instruments can certify and how many records it occupies. A *carried instrument level* is an instrument together with a level index, its own instrument record, a target list of lower levels it audits, and the ledger entries and audit records it uses. The level *occurs* if its records are carried, its ledger entries lie in Λ_S , and every index in its target list is below its own index.

The first half of E2 is imported. The *audit classifier* of Foundations III assigns to a claim about an instrument one of the verdicts accepted, undefined as circular, or outside scope.

Proposition 4.5 (E2: no same-level self-certification). *A carried self-audit claim whose subject and accepting instrument are the same level, and which has no bridge to a different level, is not accepted by the audit classifier.*

The second half counts records.

Definition 4.6 (declared capacity measure). A *declared capacity measure* for an E-system consists of

1. a depth n and a duplicate-free list of slots containing every $k < n$, with an occurring level ℓ_k of index k at each slot;
2. injective tags for instrument, ledger and audit records with pairwise disjoint ranges;
3. for each state z , a finite record universe $U(z)$ with $|U(z)| = \text{cap}(z)$;
4. for each slot, a finite set R_k that contains exactly the tags of ℓ_k 's own record, its visibility, threshold and check-rule records, and its used ledger entries and audit records; with $R_k \subseteq U(z)$ for every z , $R_j \cap R_k = \emptyset$ for $j \neq k$, and $|R_k| > 0$.

A level is *declared* if it equals some ℓ_k . The *footprint* is $\sum_{k < n} |R_k|$.

Theorem 4.7 (E2: capacity bound over a declared inventory). *For every declared capacity measure and every state z ,*

$$\sum_{k < n} |R_k| \leq \text{cap}(z).$$

If a level ℓ has the same own record as a declared level ℓ_k but an index different from k , then ℓ is not declared.

Proof. Concatenate the lists R_k over the slots. The result has no duplicates, because each R_k has none and distinct slots are disjoint, and it is contained in $U(z)$. A duplicate-free list inside a finite set has length at most the size of the set, and its length is the footprint. For the second claim, suppose $\ell = \ell_j$. Then $j \neq k$ because the indices differ, and the tag of the shared record lies in both R_k and R_j , contradicting disjointness. $\square$

Theorem 4.8 (E2: a two-level audited tower). *Extend the repair device with instrument records $\{0, 1\}$ and audit records $\{\text{yes}, \text{no}\}$. Level 0 has own record 0, which its instrument makes visible, and audits nothing. Level 1 has own record 1, visible to its instrument; it audits the lower stack $[0]$ by checking whether level 0's own record is among the records level 0's instrument makes visible, and it stores the result as the carried audit record it uses. Then:*

1. *the check returns yes, the stored audit record equals this result, and level 1 audits the whole lower stack of the depth-two tower;*
2. *with $R_0 = \{i_0\}$, $R_1 = \{i_1, a_{\text{yes}}\}$ and $U(z) = \{i_0, i_1, a_{\text{yes}}\}$, these data form a declared capacity measure, and*

$$|R_0| + |R_1| = 3 \leq 3 = \text{cap}(z);$$

3. *the device with this instrument has the lawful repair step $0 \rightarrow 1$ with audit record yes.*

Proof. The check is a finite list lookup and evaluates to *yes*. The fields of Definition 4.6 are verified by cases on the two slots: the sets are disjoint because the instrument tags are injective and instrument and audit tags have disjoint ranges. The bound is Theorem 4.7, and the repair step is checked as in Proposition 4.2. $\square$

Proposition 4.5 and Theorem 4.7 are the general statement of E2, and Theorem 4.8 is its instance. The record universe is declared to consist of exactly the three counted records, so the bound is attained. Over a declared inventory, the levels can be classified by priority — saturated if the footprint equals the capacity, otherwise rotating if an upper level audits, otherwise circular-blocked if the same-level claim is blocked, otherwise active — and this classification is exhaustive and exclusive by construction; the audited tower is saturated. That an audited repair forces a new audit level is a certificate specification: its Lean form takes the order of the level tags and the audit relation of the next level as input.

4.4 E6: priced exposure allocation

Let a probe economy (Definition 3.10) have active support P with weights $w_p \geq 0$, marginal discharges d_p and marginal costs c_p , and let spend and budget be carried ledger entries. A probe is *selected* if $p \in P$ and $w_p > 0$. A *multiplier witness* is a rational $\lambda \geq 0$ with $\mu_p \geq 0$ such that

$$d_p = \lambda c_p - \mu_p \quad (p \in P), \quad \lambda (\text{spend} - \text{budget}) = 0, \quad \mu_p w_p = 0. \quad (4)$$

These are the stationarity and complementary-slackness conditions of the problem of maximizing discharge subject to the budget [2]. The witness records supplied marginals; it names neither an objective nor a feasible set.

Theorem 4.9 (E6: multiplier algebra). *Given a multiplier witness with $c_p > 0$ for all $p \in P$:*

1. *if some selected probe has $d_p > 0$, then $\lambda > 0$, every selected probe has $d_p/c_p = \lambda$, and every $p \in P$ with $w_p = 0$ has $d_p/c_p \leq \lambda$;*
2. *if $\text{spend} < \text{budget}$, then $\lambda = 0$.*

Proof. For a selected probe $\mu_p w_p = 0$ gives $\mu_p = 0$, so $d_p = \lambda c_p$; if $d_p > 0$ and $c_p > 0$ then $\lambda > 0$. For $w_p = 0$, $d_p/c_p = \lambda - \mu_p/c_p \leq \lambda$. If spend $<$ budget, the budget slackness equation forces $\lambda = 0$. $\square$

A binding budget alone does not give $\lambda > 0$: a flat objective can spend the whole budget at price zero. The hypothesis that some selected probe has positive marginal discharge is what forces a positive price.

Proposition 4.10 (E6: when a witness is optimal). *Let a multiplier witness be given at w^* , and let*

$$F = \left\{ w \in \mathbb{R}^P : w \geq 0, \sum_{p \in P} c_p w_p \leq \text{budget} \right\},$$

with c_p the supplied marginal costs. Suppose that:

1. $D : \mathbb{R}^P \rightarrow \mathbb{R}$ is concave and differentiable;
2. the supplied marginal discharges are its partial derivatives at the allocation, $d_p = \partial_p D(w^*)$ for every $p \in P$;
3. $w^* \geq 0$ and $\sum_p c_p w_p^* \leq \text{budget}$, so $w^* \in F$; and
4. the recorded spend is $\sum_p c_p w_p^*$.

Then $D(w) \leq D(w^*)$ for every $w \in F$.

Proof. Let $w \in F$. Concavity gives $D(w) \leq D(w^*) + \sum_p d_p (w_p - w_p^*)$. By stationarity $d_p = \lambda c_p - \mu_p$, so the sum equals

$$\lambda \sum_p c_p w_p - \lambda \text{spend} - \sum_p \mu_p w_p + \sum_p \mu_p w_p^*,$$

using $\text{spend} = \sum_p c_p w_p^*$. Complementary slackness gives $\lambda \text{spend} = \lambda \text{budget}$ and $\mu_p w_p^* = 0$. Hence the sum is $\lambda(\sum_p c_p w_p - \text{budget}) - \sum_p \mu_p w_p \leq 0$, because $\lambda, \mu_p \geq 0$, $w \geq 0$ and $\sum_p c_p w_p \leq \text{budget}$. Feasibility of w^* is needed for w^* to be a candidate at all: with one probe, $D \equiv 0$, unit cost, $w^* = 2$, spend 2, budget 1 and $\lambda = \mu = 0$, every other hypothesis holds and $w^* \notin F$. No constraint qualification is needed for this direction, and positivity of the costs is not used. $\square$

Theorem 4.11 (E6: sufficiency for capped linear objectives). *Let a_0, a_1 be rational weights, c_0, c_1 rational caps and B a rational budget, and let*

$$D(x) = a_0 \min(x_0, c_0) + a_1 \min(x_1, c_1), \quad F = \{x \in \mathbb{Q}^2 : x_0, x_1 \geq 0, x_0 + x_1 \leq B\}.$$

Let $w \in F$ with $w_p \leq c_p$, and let $\lambda \geq 0$ and $\mu_0, \mu_1 \geq 0$ satisfy

$$a_p = \lambda + \mu_p, \quad \mu_p (c_p - w_p) = 0, \quad \lambda (B - w_0 - w_1) = 0 \quad (p = 0, 1).$$

Then $D(x) \leq D(w)$ for every $x \in F$.

Proof. For each p , $a_p \min(x_p, c_p) = \lambda \min(x_p, c_p) + \mu_p \min(x_p, c_p) \leq \lambda x_p + \mu_p c_p$. Summing and using $x_0 + x_1 \leq B$ and $\lambda \geq 0$,

$$D(x) \leq \lambda B + \mu_0 c_0 + \mu_1 c_1 = \lambda(w_0 + w_1) + \mu_0 w_0 + \mu_1 w_1 = a_0 w_0 + a_1 w_1 = D(w),$$

where the first equality is the two complementarity conditions and the last uses $w_p \leq c_p$. $\square$

Here μ_p is the multiplier of the cap $x_p \leq c_p$: a coordinate strictly below its cap has $\mu_p = 0$ and marginal value $a_p = \lambda$, while a coordinate at its cap may have $a_p > \lambda$. At a cap, 0 is a valid supporting slope of D in coordinate p , and with that choice the multiplier of the cap, not the budget price alone, carries the difference. The theorem states sufficient conditions; a coordinate at zero below its cap is required to satisfy $a_p = \lambda$, which is stronger than necessary.

Theorem 4.12 (E6: an optimum at a cap). *With weights $a = (2, 1)$, caps $c = (1, 5)$ and budget $B = 3$, the allocation $w = (1, 2)$ is optimal: $D(x) \leq D(w) = 4$ for every $x \in F$. The multipliers $\lambda = 1$, $\mu = (1, 0)$ satisfy the conditions of Theorem 4.11, the first coordinate sits at its cap with $\mu_0 = 1 > 0$, and its supporting slope is 0. With unit weights, $w = (1, 2)$ is also optimal.*

Proof. $a_0 = 2 = 1 + 1$, $a_1 = 1 = 1 + 0$, $\mu_0(c_0 - w_0) = 1 \cdot 0$, $\mu_1(c_1 - w_1) = 0 \cdot 3$ and $\lambda(B - w_0 - w_1) = 1 \cdot 0$; apply Theorem 4.11. With unit weights, $\min(x_0, 1) + \min(x_1, 5) \leq x_0 + x_1 \leq 3 = D(w)$. $\square$

Theorems 4.9 and 4.11 and Proposition 4.10 are the general statement of E6, and Theorem 4.12 is a numerical instance of Theorem 4.11: its allocation and multipliers are rationals, not records carried in the ledger of an E-system. An instance of E6 on a carried probe economy is open. The two examples below show that neither concavity nor the link between supplied marginals and the objective can be dropped from Proposition 4.10.

Example 4.13 (multipliers without concavity). Maximize $D(x, y) = x^2 + y^2 + x + y$ over $x, y \geq 0$ with $x + y \leq 1$, with unit costs. At $(1/2, 1/2)$ both marginal discharges equal $2x + 1 = 2$, so $\lambda = 2$, $\mu = 0$ satisfy (4) with a binding budget and a selected probe of positive discharge. Yet $D(1/2, 1/2) = 3/2 < 2 = D(1, 0)$. This example is argued on paper.

Remark 4.14 (multipliers not linked to the objective). On the unit simplex take $D(x) = x_0$, supplied marginals $d = (1, 1)$, unit costs and $x^* = (0, 1)$ with spend = budget = 1. Then $\lambda = 1$ and $\mu = 0$ satisfy (4) and probe 1 is selected with $d_1 > 0$, yet $D(x^*) = 0 < 1 = D(1, 0)$: hypothesis (2) of Proposition 4.10 fails. This remark is argued on paper.

The name “attention” is operational; the law says nothing about awareness.

4.5 E1: perpetual self-extension

E1 asks when a recurring challenge forces a system to extend itself again and again. A *repair family* assigns to some times t a history entry and an installed refinement $R_t : X \rightarrow \mathcal{R}$. A *novelty predicate* N says, for a challenge class $\mathcal{C}$ and the system’s state of knowledge σ_t at time t , whether the challenge is new at t ; a *strictness predicate* P says whether a refinement is a strict extension relative to σ_t . The family makes *infinitely many strict self-extensions* if for every n there is a time $t \geq n$ with an entry whose refinement satisfies $P(R_t, \sigma_t)$.

Definition 4.15 (actual-family forcing). An *actual-family forcing certificate* for a repair family consists of

1. *recurrence*: for every n the family has an entry at some time $t \geq n$; and
2. *installation*: at every entry time t at which $N(\mathcal{C}, \sigma_t)$ holds, the entry’s refinement satisfies $P(R_t, \sigma_t)$.

Proposition 4.16 (E1: forced self-extension). *If a repair family has an actual-family forcing certificate and the challenge is new at every time, then the family makes infinitely many strict self-extensions.*

Proof. Given n , recurrence gives an entry at some $t \geq n$, and installation applied to the novelty at t makes its refinement strict. $\square$

The proposition is a direct combination of the two fields; its content lies in the instance, where recurrence and installation are proved about executed repairs.

Theorem 4.17 (E1: a clocked device). *Let the clocked device have carrier $\mathbb{N}$, kernel $z \rightarrow z + 1$ and trajectory $\tau(t) = t$. A step from an even to an odd state is a perturbation, and at an odd state t the device carries the defect record t , its generator produces a move whose payload is t , and the move is admissible exactly at odd states. Let the family have an entry exactly at the odd times t , with the move entry read at time $t + 1$ and the refinement $R_t(b) = t + 1$ if b is true and $R_t(b) = t$ otherwise. Let $\sigma_t = (t, \{0, \dots, t - 1\})$, let N say that $t \notin \{0, \dots, t - 1\}$, and let $P(R, \sigma_t)$ say that $R(\text{false}) \notin \{0, \dots, t - 1\}$ and $R(\text{true}) \neq R(\text{false})$. Then:*

1. *at every odd t , the step $t \rightarrow t + 1$ is a lawful repair step with defect t , the move payload t installs R_t , and the move entry is carried;*
2. *every entry of the family is of this form, and the family has entries beyond every bound;*
3. *the family has an actual-family forcing certificate, the challenge is new at every time, and so the family makes infinitely many strict self-extensions;*
4. *refinements installed at distinct times are distinct.*

Proof. Part (1) checks the repair step at each odd time; parts (2) and (3) read recurrence, installation and novelty off the construction and apply Proposition 4.16; part (4) compares refinements. (1) At odd t the instrument detects the defect t , the gate and admissibility hold, the ledger line is the single entry, $(t, t + 1)$ is a kernel step and the re-audit passes at the even state $t + 1$; the move payload is the defect t , and the installation rule sends payload t to R_t . (2) The family is defined by parity, and $2n + 1 \geq n$ is odd. (3) Recurrence is (2); installation holds because $R_t(\text{false}) = t \notin \{0, \dots, t - 1\}$ and $t + 1 \neq t$; novelty is $t \notin \{0, \dots, t - 1\}$. Proposition 4.16 gives the conclusion. (4) $R_t(\text{false}) = t$. $\square$

The instance is weak in the sense of Section 3.2: novelty is the fact that a clock value has not occurred before, and strictness is a chosen fresh output, so the theorem shows that the certificate can be met by executed, carried repairs with pairwise distinct refinements, not that a system facing an open-ended challenge must extend itself.

4.6 E12: individuation

E12 asks which parts of a system are individuals: parts that keep their repairs inside themselves, maintain their own boundary, and have no rival claim to the same states. We state it for a finite system directly.

Definition 4.18 (individuation setting). An *individuation setting* consists of a finite set of states Z , a step relation $\rightarrow$ on Z , a closure operator cl on subsets of Z (extensive, monotone and idempotent), and a finite list of *boundary regions* $B_1, \dots, B_m \subseteq Z$. A *part* is a subset $P \subseteq Z$. A part is

1. *repair-closed* if $\text{cl}(P) = P$ and every step from a state of P ends in P ;
2. *self-maintaining* if it is nonempty, lies inside some boundary region, and every state of P has a step to a state of P ;
3. a *candidate* if it is both;
4. *maximal* if it is a candidate and no strictly larger part is a candidate;

5. an *integrated individual* if it is maximal and every maximal part that overlaps it equals it.

The closure operator says which states belong together; in the instance below it joins each healthy state with the damaged state it is perturbed into and repaired from. The boundary regions are the parts whose boundary the system can hold; a part outside every region cannot maintain itself, however it behaves.

Proposition 4.19 (E12: structure of candidates). *In every individuation setting:*

1. *the intersection of two repair-closed parts is repair-closed;*
2. *two integrated individuals are equal or disjoint.*

Proof. Part (1) uses monotonicity and extensivity of the closure, and part (2) uses the overlap condition that defines an integrated individual. (1) Monotonicity gives $\text{cl}(P \cap Q) \subseteq \text{cl}(P) \cap \text{cl}(Q) = P \cap Q$, and extensivity gives the reverse inclusion; a step from $P \cap Q$ stays in P and in Q . (2) If they overlap, each is a maximal part overlapping the other, so they are equal by definition. $\square$

Part (2) is immediate from the definition; the content of E12 is in the instances.

Theorem 4.20 (E12: an individual that contains its own repair). *Let $Z = \{h, d, o\}$ (healthy, damaged, outside). The steps are the stutters $z \rightarrow z$ and the repair $d \rightarrow h$; a perturbation sends h to d but is not a step of the system. The closure adds d to any part containing h and h to any part containing d , and fixes o . The single boundary region is $\{h, d\}$. Then:*

1. *$\{h, d\}$ is an integrated individual; it contains the damaged state together with its repair, and it is neither empty nor all of Z ;*
2. *$\{h\}$ is closed under steps and self-maintaining, but it is not a candidate, because its closure contains d .*

Proof. Both parts are checked directly from the closure, the steps and the single boundary region. (1) The closure of $\{h, d\}$ is itself, the only non-stutter step $d \rightarrow h$ stays inside, the part lies in the boundary region, and every state has its stutter. Every candidate lies in the single region $\{h, d\}$, so no candidate is strictly larger and every overlapping maximal part is contained in, hence equal to, $\{h, d\}$. The state o is outside it. (2) h has only its stutter, and $\{h\} \subseteq \{h, d\}$; but $\text{cl}(\{h\}) = \{h, d\} \neq \{h\}$. $\square$

Part (2) shows that the closure condition does work: the healthy state alone is closed under the system's steps and maintains itself, and only the closure, which joins it to the damaged state, excludes it.

Proposition 4.21 (E12: overlapping maxima leave no individual). *Let $Z = \{0, 1, 2\}$, with only stutter steps, the identity closure and boundary regions $\{0, 1\}$ and $\{1, 2\}$. Both regions are maximal parts, they overlap at 1, and no integrated individual exists.*

Proof. Each region is a candidate; any candidate lies in one region, so a candidate containing a region is that region. An integrated individual would lie in one region, hence equal it by maximality, and then the other region would be an overlapping maximal part different from it. $\square$

The setting models repair closure, boundary maintenance and the exclusion of rivals. It does not model the budget and record closure of the full classification (Template S4.10), and a boundary region is a declared set, not a quotient shown to be the coarsest one through which the viability probes descend.

4.7 E13: repair transport

E13 asks when a repair in one system becomes a repair in another: when a signal that works for the sender also reduces the same kind of difficulty for the receiver.

Definition 4.22 (repair system and transport). A *repair system* has states S , contexts C , a repair relation $s \rightarrow_c t$ in each context, an obstruction $\text{ob}_c(s) \in \mathbb{N}$, a challenge class $\text{cls}(c)$ for each context and a role $\text{role}(s)$ for each state. A *paid interface* from A to B consists of maps f on states and g on contexts that preserve roles and challenge classes, and a payment $\pi > 0$. A *repair transport* is a paid interface such that for every source repair $s \rightarrow_c t$ in A ,

$$f(s) \rightarrow_{g(c)} f(t) \text{ in } B \quad \text{and} \quad \text{ob}_{g(c)}^B(f(t)) < \text{ob}_{g(c)}^B(f(s)).$$

Theorem 4.23 (E13: transports compose and bound repair chains). 1. *Repair transports $A \rightarrow B$ and $B \rightarrow C$ compose to a repair transport $A \rightarrow C$ whose payment is the sum of the two payments.*

2. *If $s_0 \rightarrow_c s_1 \rightarrow_c \dots \rightarrow_c s_k$ is a chain of k repairs in A and (f, g) is a repair transport to B , then*

$$\text{ob}_{g(c)}^B(f(s_k)) + k \leq \text{ob}_{g(c)}^B(f(s_0)),$$

so no chain of source repairs is longer than the receiver's obstruction at its start.

Proof. (1) Compose the maps; roles, classes, mapped repairs and strict decrease pass through both steps. The payment of the composite is defined as the sum. (2) Induction on k : each transported repair lowers a natural-number obstruction by at least one. $\square$

Theorem 4.24 (E13: transport across contexts). *Let the source have states $\mathbb{N}$, contexts $\{0, 1\}$ of class 7, repairs $s \rightarrow_c t$ for $t < s$, obstruction $s + c$ and role $s \bmod 2$. Let the receiver have states $\mathbb{N} \times \{\text{true}, \text{false}\}$, the same contexts and class, repairs that lower the first coordinate and keep the second, obstruction $s_1 + c$ and role $s_1 \bmod 2$. Let $f(s) = (s, \text{true})$, let g exchange the two contexts, and let the payment be 4. Then (f, g) is a repair transport, and the two-step chain $3 \rightarrow_0 2 \rightarrow_0 1$ lowers the receiver's obstruction in context 1 by at least 2. Composing the identity transports to the receivers with obstruction $s + c + 1$ and $s + c + 2$, with payments 2 and 3, gives a transport with payment 5 that reduces obstruction in both contexts.*

Proof. Direct verification, then Theorem 4.23. $\square$

Proposition 4.25 (E13: a paid, role-preserving map need not transport repair). *The identity map from the source of Theorem 4.24 to a receiver with the same repairs, roles and classes but constant obstruction 0 is a paid interface that carries every source repair to a receiver repair, yet reduces no obstruction.*

Proof. The maps preserve roles and classes, and a source repair $t < s$ is a receiver repair; but the source repair $2 \rightarrow_0 1$ would need $0 < 0$. $\square$

The theorems concern paid, role-preserving repair transport, the core of the communication case of E13. The full communication certificate, which adds carried evidence and mediation by an interface quotient, and the teaching, coercion and symbolic cases (Template 6.2) remain certificate specifications.

5 General theorems

The results of this section have ordinary hypotheses and nothing open. Together with Propositions 3.3 and 3.5 of Section 3.3 and Proposition 5.4 of Section 6, they are the general theorems of the catalog.

5.1 E9: strict probe acquisition

Allocation reweights probes the system already has; acquisition adds a new one. For an active family L , a catalog probe M is an *admissible acquisition* if it lies in the catalog and outside the support of L , the acquisition is strict (Definition 3.10), and a carried budget record makes it feasible and within the declared viability-risk bound. Let $\delta(M)$ and $c(M)$ be supplied rational discharge and cost functions and write $\delta(M)/c(M)$ for the ratio.

Theorem 5.1 (E9: finite argmax). *Let a finite nonempty list of candidates consist of admissible acquisitions and contain every admissible acquisition. Then some listed M^* is admissible and satisfies $\delta(M)/c(M) \leq \delta(M^*)/c(M^*)$ for every admissible acquisition M . The same holds for a list of access moves that mixes allocation increments and acquisitions under a common ratio. If a policy selects a listed move that maximizes the ratio over the list, that move dominates every admissible alternative.*

Proof. A finite nonempty list of rationals has a maximum; completeness extends the comparison from the listed candidates to all admissible ones. The statement about a policy is the same extension applied to the selected move. $\square$

Proposition 5.2 (saturated probes discharge nothing). *In the operator setting let $\delta(M) = \text{tr}(K_{DM|L} K_{MM|L}^\dagger K_{MD|L})$. If the chain rule (3) holds, $M = BL$ and $K_{DM|L} = 0$, then $\delta(M) = 0$.*

Proof. The trace is of a product whose first factor is zero. $\square$

With the operator discharge and positive costs, the maximizer of Theorem 5.1 has the form

$$M^* \in \arg \max_M \frac{\text{tr}(K_{DM|L} K_{MM|L}^\dagger K_{MD|L})}{c(M)}$$

over admissible acquisitions. The theorem maximizes whatever ratio is supplied; identifying the discharge with the operator contraction is a specialization made by the application. Ties are allowed, and the name “curiosity” is operational.

5.2 E15: the budget-reallocation identity

E15 concerns systems that alternate between an online phase, in which part of a shared budget is spent on external exchange, and an offline phase, in which exchange is gated off and the freed budget is used for internal discharge of closure debt.

Proposition 5.3 (offline capacity is the freed allocation). *Let a carried allocation record assign, in the online phase, an external share e and an internal share $\kappa_{\text{on}} \geq 0$, and in the offline phase an external share 0 and an internal share κ_{off} . Suppose the shared budget B is binding and exhausted in both phases: $e + \kappa_{\text{on}} = B = 0 + \kappa_{\text{off}}$. Then*

$$\kappa_{\text{off}} = \kappa_{\text{on}} + e,$$

and $\kappa_{\text{on}} < \kappa_{\text{off}}$ if $e > 0$.

Proof. Subtract the two budget equations. $\square$

The identity says where the offline capacity comes from; whether an offline phase is needed at all is the subject of Template S4.13.

5.3 E4: brittleness of compiled repair

A repair that has been compiled into lower support is tuned to the class it was trained on.

Proposition 5.4 (E4: brittleness). *Let a compiled repair induce a quotient q on the trained challenge class C , let q^b be the minimal quotient generated from q for the readout on C , and let $C' \neq C$. If (x, x') is a split pair for the update $F_{C'}$ through q^b , then the readout of $F_{C'}$ does not descend through q^b .*

Proof. Descent is equivalent to the absence of split pairs [26]. □

The mis-compiled status of Template S4.2 uses its own out-of-class obstruction witness; it is not formally linked to the split pair of Proposition 5.4.

6 Certificate specifications

The remaining entries are certificate specifications (Section 3.2), except E5 and E8, which are definitions with consistency lemmas. Each certificate specification has a Lean statement of the form “a complete certificate yields the classification”: the certificate is a carried status record asserting that one branch of the classification holds for a claim, and Lean proves that the branches are mutually exclusive and derives the fields the branch requires. Supplying the certificate for a real system is the open obligation. In every case the certificate is a complete, branch-valid status record; Table 4 names, for each entry, the evidence its principal branch requires. This section prints two in full, E3 and E13, which complement the laws of Section 4. The others are printed, with meanings and examples, in Supplement Section S4, and their full field lists are in Supplement Section S5.

Table 4. Certificate specifications and definitions.

Entry	Classification	Evidence for the principal branch
E1 internalization	slack, externally subsidized, endogenously repairing, stressed, collapsing	endogenously repairing: a binding challenge, the repair family, and no discharging entry of the claim with a forbidden source, missing generation, scope or carriedness, an off-kernel move or an omitted audit (Template S4.1)
E4 repair compilation	exposed, compiling, compiled, mis-compiled, decompiled	compiled: a compilation case in which the repeated repair enters lower support (Template S4.2)
E5 reclosure collapse	seven statuses, among them collapsed-recoverable, collapsed-irreversible and revived	collapsed: a complete inventory of rescue candidates, none admissible within budget and viability-descending (Definition S4.3)
E7 alarm	currentization, boundary nonclosure, lawful discount	a unique total disposition for the specified viability-coupled blind spot and horizon (Template S4.5)
E8 control price	seven statuses of a control signal	component-lawful: the shadow-price components of a carried control signal (Definition S4.6)

Entry	Classification	Evidence for the principal branch
E10 cognitive demarcation, intention, goal	eight, five and four statuses	cognitive, intention, goal: status records and comparators that exclude schedule traps, post-hoc intentions and reward proxies (Template S4.8)
E11 institutional rewrite	inert, conditioning, stack-active, constitutive	constitutive: structural evidence of a changed lower layer and a separate compilation certificate (Template S4.9)
E12 individuation, full classification	integrated, federated, subsidiary, platform-dependent, shadow, non-individuated	integrated: repair, budget and record closure, a coarsest viability boundary, maximality and no overlapping rival (Template S4.10)
E14 reconsolidation	nine statuses; the lawful ones are record repaired, record coarsened and statused unresolved	a lawful disposition: complete inventories of repair, coarsening and statused outcomes, provenance for every mutation, and the exclusion of the defect statuses (Template S4.12)
E15 offline reclosure	eight statuses, among them alternation required and online sufficient	alternation required: a strict online deficit, recurrent offline phases with exchange gated to zero and linked debt discharge, and a duty cycle within a declared tolerance (Template S4.13)
E16 adaptability	eight statuses, among them coherent adaptability and flat	coherent adaptability: a record that two routes agree now and differ in future repair capacity, with its controls (Template S4.14)

6.1 E3: maintained closure

Template 6.1 (E3: maintained closure). *Data.* A running package with apparatus App_t , a maintenance operator $\mathbf{m}$ and a reinstatement record. *Certificate.* A complete maintenance status record with four statuses: crystal-grade, maintained, subsidized and decaying closure. The maintained branch contains an object fixed point and record coherence, a carried maintenance operator, the reinstatement condition of Section 4.2, a maintained step (apparatus distance at most ε without challenge, a linked lawful repair step under challenge), an empty maintenance obstruction and no external reinstatement witness. *Conclusion.* The four statuses are mutually exclusive, and the maintained branch yields both object closure and apparatus reinstatement.

Meaning. A persisting object is classified by who keeps its apparatus in place. *Crystal-grade* closure is persistence with a stable apparatus and no maintenance operator or reinstatement at all, as in a crystal; *maintained* closure has the system’s own reinstatement; *subsidized* closure has an external or fallback reinstatement; *decaying* closure has apparatus drift above tolerance and no valid reinstatement. The maintenance obstruction is nonempty if a required apparatus component is absent, has a forbidden source, fails generation or scope, is off-kernel, or is omitted from the declared audit. The history-linked form of reinstatement (Definition 4.3) is the one with a proved instance. A further Lean statement packages measured apparatus and object distance curves after the maintenance channel is cut. *Example.* After the maintenance channel is cut, the apparatus

distance runs $0, 1/8, 3/8, 1/2, 5/8, 3/4$ and the object distance $0, 1/16, 1/8, 3/16, 5/16$; they separate first at time 2. A crystal-grade control never separates.

6.2 E13: repair transport

Template 6.2 (E13: repair transport). *Data. E-systems A and B joined by an audited, paid bridge; a token s; a challenge class C shared by source and receiver records. Certificate. One of four full cases, with a complete status record over coercion null, symbolic, taught, communicated, scaffolded-or-primed, influence-only and transport-rejected:*

1. communication (E13.1): *s induces a paid, interface-mediated, role-preserving carried receiver repair with a strict reduction of B's obstruction in class C;*
2. teaching (E13.2): *communication, and at a later probe with A absent and the bridge closed, the repair is carried by B from its own committed state;*
3. coercion null (E13.3): *a force comparator classifies the interaction as fully forced, a carried free-response census is empty, no B-committed transported repair of the token exists and the obstruction is unchanged;*
4. symbolic repair (E13.4): *s reactivates a carried, role-preserving, strict repair in every context of a declared family of at least two contexts.*

Conclusion. The corresponding judgment holds, and its case records the absence of raw evidence for every higher-priority judgment of the same claim; coercion-null evidence excludes all six lower judgments.

Meaning. A communicated token leaves the receiver with a carried repair that strictly reduces its obstruction in the shared class, through a paid, interface-mediated, role-preserving bridge; teaching adds that the repair is carried by the receiver from its own committed state when the teacher is absent and the bridge is closed. Paid, role-preserving repair transport, the core of the communication case, is the law of Section 4.7; its instance transports repair across two distinct contexts of one challenge class. The full communication certificate and the teaching, coercion-null and symbolic cases remain certificate specifications. *Example.* An interaction that a force comparator classifies as fully forced, with a carried free-response census that is empty, no receiver-committed transported repair of the token, and an unchanged obstruction, is a coercion null.

7 Scope

7.1 What the results do not give

Each entry is easy to read more strongly than it is stated. Table 5 lists the most tempting inferences and what blocks each one. Further ones are listed in Section S6.1, and the full register of non-implications, with a separating example for each, is in Supplement Section S6.

Table 5. Tempting inferences and what blocks them.

Tempting inference	Why it fails
A repair join is a repair.	The join supplies a common refinement; whether a repair occurs, who produced it and whether it reduces an obstruction are separate (Proposition 3.3).

Tempting inference	Why it fails
A carried record is a true record.	Carriedness concerns the source and trajectory of a record, not its truth or completeness (Definition 3.6).
A closed-loop history shows self-repair.	A history with no repair entries is closed-loop (Lemma S3.3).
One lawful repair step shows maintenance.	The device takes one lawful step; recurrence, reduction of an obstruction and maintenance are further certificates (Lemma S3.2).
The audited tower shows that audit towers are small.	Its record universe is declared to be exactly the three counted records, so the bound is attained by construction (Theorem 4.8).
A multiplier witness identifies an optimal allocation.	Only under the hypotheses of Proposition 4.10 or Theorem 4.11; Example 4.13 and Remark 4.14 satisfy the multiplier conditions and are not optimal.
A binding budget gives a positive price.	A flat objective spends the whole budget at price zero; a selected probe with positive discharge is needed (Theorem 4.9).
The clocked device shows that an open-ended challenge forces self-extension.	Its novelty is a fresh clock value and its strictness a chosen fresh output (Theorem 4.17).
A paid, role-preserving map transports repair.	Not unless it strictly lowers the receiver’s obstruction (Proposition 4.25).

7.2 Domain readings

A domain name does not instantiate an entry. Table 6 lists, for each entry, candidate readings in living, cognitive and social systems, and the data an application must supply before a reading becomes an instance. The readings are candidates for testing; none is claimed as an instance here. Every application must in addition answer the common questions: which finite states or histories are compared and at what scale; which quotients are current, predictive or target-side, with a literal split pair for each failure of descent; which instrument, ledger, repair, boundary and probe records are coordinates of the carrier and updated by its kernel; which resource is charged and in what units; which source produced each credited record and which external, fallback, off-kernel or wrong-lineage records are excluded; and which controls were run. Three guards apply throughout. Alarm is preemption, curiosity is strict acquisition and goal is persistent target selection; the words attribute no feeling or purpose. A ledger may be energetic, material, computational, financial or legal, and calling it a budget does not make these convertible. Individual, lineage, population, organization and ecosystem are different carrier choices, and evidence at one does not establish carriedness at another.

Table 6. Domain readings and the data each requires.

Entry	Living systems	Cognitive systems	Social systems	An instance must supply
E1	wound closure, immune response, regeneration	policy and abstraction repair	amendment, crisis playbooks	a repair record produced by the carrier, not the experimenter; the complete audited history
E2	molecular checkpoints, tissue surveillance	metacognitive monitors	audit committees, regulators	explicit audit scopes and a declared record inventory with capacity

Entry	Living systems	Cognitive systems	Social systems	An instance must supply
E3	membrane turnover, gradient and catalyst restoration	calibration, memory integrity	renewal of staff, capital and procedures	a carried maintenance operator and a history-linked reinstatement; an ablation of the operator
E4	immune memory, canalization	skill and policy compilation	precedent, standard contracts	matched interventions showing a lower-support rewrite; a memory-only control
E5	stress, dormancy, life support, irreversible loss	a severed learning channel	illiquidity, receivership, bankruptcy	viability probes and the complete rescue inventory
E6	receptor expression, immune surveillance	selective attention, active sensing	inspection and monitoring allocation	costs and residual discharge in common units; a concave objective or a direct optimality argument
E7	DNA-damage checkpoints, danger signals	interrupt-driven replanning	margin calls, circuit breakers	viability coupling of the witness and a changed constraint set
E8	homeostatic and allostatic signals	compute and error control	transfer prices, capital adequacy	a binding constraint, its multiplier, lineage audit and slack control
E9	receptor repertoire change	active learning, queries	research, audit sampling	a complete candidate list, positive costs, strictness against reweighting
E10	regulatory packages that gate action	plan-gated action versus decodable state	decision procedures versus minutes	repair descent, a certified gate and carried provenance for the same package
E11	developmental support rewrite	plan-to-execution rewrite	constitutional or platform rewrite	matched upper interventions with a support or closure difference
E12	cell, organism, colony, symbiosis	modular agent	firm, subsidiary, federation	all five closure and boundary tests and the candidate list
E13	signalling that transfers repair capacity	instruction	teaching, standards, law	carried transport records, paid interface, receiver-side reduction
E14	epigenetic and immune record revision	reconsolidation after retrieval	restatement, versioned archives	the retrieval conflict and complete outcome inventories
E15	quiescence, repair phases	replay and consolidation	settlement and maintenance windows	debt measure, shared budget allocation, duty-cycle prediction
E16	priming, trained immunity	curriculum-order transfer	crisis experience	two routes with matched endpoints and the full control battery

The atlas shows two things. Carried repair, audit capacity, priced sensing and apparatus maintenance are biological from the first entries on; life does not first appear at E12 or E16. And the cognitive column is not a promotion of the living one: a living system may or may not meet the E10 criterion, and every other entry is meaningful either way.

7.3 What the catalog leaves out

Several candidates have a recognizable shape but no distinct obstruction or theorem yet: global or conscious access, which may combine E6, E7, E12 and E13; play, which currently reads as E9 under a gated risk budget; the search for boundaries, which E12 classifies but does not produce; transport

of gates and norms rather than repairs; distributed price discovery across carriers; reproduction, heredity and population selection; and the origin of a first E-system from non-living chemistry. The catalog defines neither life nor death, pathology or welfare.

7.4 Open problems

The sorting of Table 3 suggests the following problems, each of which would move an entry to a stronger status.

1. *Stratification from a construction rule.* Derive the order of the level tags and the next level’s audit relation of E2 from a rule for adding a level, rather than supplying them, and restate the regress stop of E3 and the meta-audit bound of E8.1 over the declared measure.
2. *Richer instances.* Replace the single-valued record sorts and the declared completeness propositions of the repair device by a model in which defects, inventories and formedness have content, so that the history link of Theorem 4.4 and the audit of Theorem 4.8 carry information.
3. *Optimality beyond capped linear objectives.* Extend Theorem 4.11 to n probes, general costs and a wider class of concave discharge objectives, and derive the spend of the ledger from the objective.
4. *Open-ended novelty.* Give an instance of Proposition 4.16 in which novelty and strictness are measured on the obstruction the challenge causes, rather than on a clock.
5. *Individuals and transport.* Add budget and record closure and a boundary derived as the coarsest viability-sufficient quotient to Definition 4.18; prove that a maximal candidate exists above every candidate; and add carried sending and receiving records to Definition 4.22, with an instance of teaching.
6. *Route effects.* Supply an intervention bridge under which the unequal repair probabilities of E16 are an effect of the route.

8 Formalization

The definitions D1–D6, the general statements of E1–E16 and the laws and instances of Section 4 are checked in Lean 4 [6], in a library with no external dependencies beyond the vendored interfaces of the earlier papers of the series. The library builds without `sorry` and declares no axioms. Following Pollack [16], what a reader must trust is the kernel and the statements; Supplement Section S2 maps every numbered result of this paper to its Lean declarations, file and line, so that each statement can be compared with the printed one.

Axioms. Every declaration uses at most the three standard axioms `propext`, `Quot.sound` and `Classical.choice`; the record for each theorem is in Supplement Section S9.

What each checked statement covers.

- Definitions: Propositions 3.3 and 3.5 and Lemmas S3.1 to S3.4. In Proposition 3.5 the chain rule and positive semidefiniteness are hypotheses.
- Laws and instances: Propositions 4.2, 4.5, 4.16, 4.19, 4.21 and 4.25 and Theorems 4.4, 4.7, 4.8, 4.17, 4.20, 4.23 and 4.24. The converse of Theorem 4.4 is checked for the form of the definition that requires only the move entry. Proposition 4.5 is checked in the weaker form that the classifier does not accept the claim. In Theorem 4.23 the payment of the composite is the sum by definition.

- General theorems: Theorems 4.9, 4.11 and 5.1 and Propositions 5.2 to 5.4, with the numerical instance Theorem 4.12 of Theorem 4.11. The identity of Proposition 5.3 is derived from the two exhaustion equations and the gate.
- Certificate specifications and definitions: the Templates of Section 6 and Supplement Section S4, Proposition S4.11, and Lemmas S4.4 and S4.7, each exactly as printed: the certificate is a hypothesis, and the conclusion is derived from it.

Argued on paper only. Proposition 4.10, Example 4.13, and Remark 4.14; the envelope reading of E8; alarm fatigue (E7.1); and operational selfhood (E12.1).

Declared propositions. Formedness of a package and completeness of record, ledger and history inventories are propositions supplied with the data. Lean checks what follows from them, not that they hold of a real system. In the repair device and the clocked device they are declared true, which is why their instances are weak. The instance of E6 is numerical and carries no ledger records.

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A Revision notes

This appendix lists the changes from the first version of the paper. The main text states the results as they now stand and does not refer to the earlier version.

Proposition A.1 (the version-1 capacity measure is vacuous). *Version 1 measured an audit tower by a record universe with $|U(z)| = \text{cap}(z)$ and, for every occurring level ℓ , a record set $R(\ell) \subseteq U(z)$ containing exactly ℓ 's counted records, with $R(\ell) \cap R(\ell') = \emptyset$ whenever ℓ and ℓ' occur and have different indices. No E-system with an occurring level admits such a measure.*

Proof. Let ℓ occur with index k , and let ℓ' be ℓ with index $k + 1$ and an empty target list. Then ℓ' occurs, and the tag of the common own record lies in $R(\ell)$ and in $R(\ell')$, contradicting disjointness. $\square$

Proposition A.2 (the version-1 forcing certificate is uninhabited). *Version 1 derived infinitely many strict self-extensions from a finite-forcing certificate that asserts the conclusion for every repair family, every novelty predicate and every strictness predicate whenever the challenge is new at every time. No system with inhabited state types admits such a certificate.*

Proof. Take the novelty predicate that always holds and the empty family. The certificate asserts that the empty family has an entry beyond every bound, and it has none. $\square$

1. *E1.* By Proposition A.2, the strict-self-extension statement of version 1 had no instance. The actual-family certificate of Definition 4.15 replaces it, with the clocked instance of Theorem 4.17; E1 is now a law.
2. *E2.* By Proposition A.1, the capacity bound of version 1 held only for systems without an occurring instrument level. The same measure was a hypothesis of forced stratification, of the level classification, of the regress stop of E3 and of the meta-audit bound of E8.1; none of these applies to a system with an occurring level. The declared measure of Definition 4.6 replaces it, with the instance of Theorem 4.8. The version-1 declarations remain in the library.
3. *E3.* The reinstatement condition of version 1 names a challenge history but does not use it, so it holds with an empty history. The history-linked form of Definition 4.3 replaces it, and the device now executes its repair along its declared trajectory.
4. *E6.* Version 1 called the witnessed allocation “certified optimal”. The multiplier witness yields the consequences of Theorem 4.9, not optimality; optimality holds under the hypotheses of Proposition 4.10, and is checked in Lean for capped linear objectives (Theorem 4.11).
5. *E12 and E13.* In version 1 both were templates whose certificates had budget, viability and completeness fields that no concrete model could establish. They are now laws with the semantic definitions of Definitions 4.18 and 4.22; the version-1 Templates are kept as the full classifications.
6. *E15.* The budget-reallocation identity was a field of the allocation record in version 1; it is now derived from the exhaustion equations.
7. *Presentation.* The paper leads with the laws and their instances, gives the certificate specifications afterwards, and moves the concordance, field lists and registers to the supplement.